


```
Python
#import arcpy and set workspace environment
import arcpy
arcpy.env.workspace = "C:\\GEOS456\\Data.gdb"
#allow the script to overwrite existing outputs
arcpy.env.overwriteOutput = True
#call the buffer tool and fill in required parameters
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer", "100 Meters")
#use "GetMessages()" to return all the tool messages
print (arcpy.GetMessages())
```

6. Examine the results in the Interpreter

```
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#import arcpy and set workspace environment
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#call the buffer tool and fill in required parameters
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer", "100 Meters")
#use "GetMessages()" to return all the tool messages
print (arcpy.GetMessages())
Start Time: Tuesday, August 17, 2021 2:00:16 PM
Succeeded at Tuesday, August 17, 2021 2:00:22 PM (Elapsed Time: 6.18 seconds)
```

The above screenshot shows the result of the tool. Using the **GetMessages()** function returns the same information as when a tool is run and shown in the results window in ArcMap.

7. Remove the **GetMessages()** line and replace the last line of the script with the **GetMessage()** function

8. Run the script

```
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Succeeded at Tuesday, August 17, 2021 2:00:22 PM (Elapsed Time: 6.18 seconds)
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#call the buffer tool and fill in required parameters
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer", "100 Meters")
#use "GetMessages()" to return all the tool messages
print (arcpy.GetMessage(0))
Start Time: Tuesday, August 17, 2021 2:02:05 PM
```

The only message that returned is the first line of the message when a tool is run. This is because the **GetMessage()** function used the index value of **0**. Remember that indexing in Python always starts at zero and zero will always return the first value.

9. Remove the **GetMessage(0)** line and replace the last line of the script with the **GetMessageCount()** function

```
Python
#import arcpy and set workspace environment
import arcpy
arcpy.env.workspace = "C:\\GEOS456\\Data.gdb"
#allow the script to overwrite existing outputs
arcpy.env.overwriteOutput = True
#call the buffer tool and fill in required parameters
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer", "100 Meters")
#use "GetMessageCount()" to return the LAST tool message
count = arcpy.GetMessageCount()
print (arcpy.GetMessage(count -1))
```

NOTE: The `GetMessageCount()` function does not accept a parameter in itself. This is why you must set a variable, in this case “count”, and then reference that variable with an index value using the `GetMessage()` function.

10. Run the script

```
Python
#import arcpy and set workspace environment
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arcpy.env.workspace = "C:\\GEOS456\\Data.gdb"
#allow the script to overwrite existing outputs
arcpy.env.overwriteOutput = True
#call the buffer tool and fill in required parameters
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer", "100 Meters")
#use "GetMessageCount()" to return the LAST tool message
count = arcpy.GetMessageCount()
print (arcpy.GetMessage(count -1))
Succeeded at Tuesday, August 17, 2021 2:11:44 PM (Elapsed Time: 6.25 seconds)
```

The only message that returned is the last line of the message when a tool is run. This is because the ***GetMessage()*** function used the index value of ***-1***. Remember that indexing can also work in reverse. Python uses an index value of ***-1*** to retrieve the last value.